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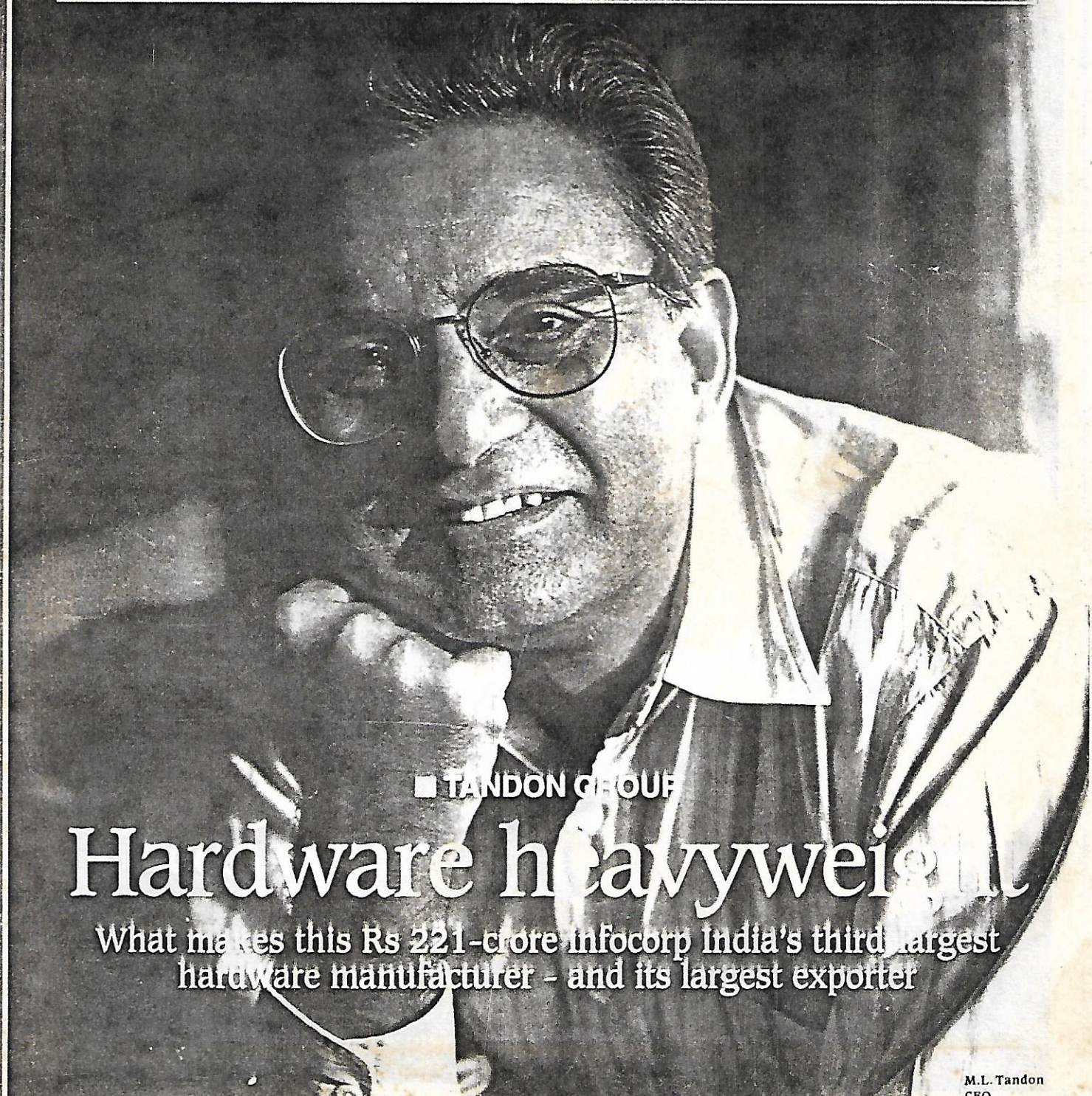
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■ TANDON GROUP

Hardware heavyweight

What makes this Rs 221-crore Infocorp India's third largest hardware manufacturer - and its largest exporter

M.L. Tandon
CEO



Hardware h[illegible]t

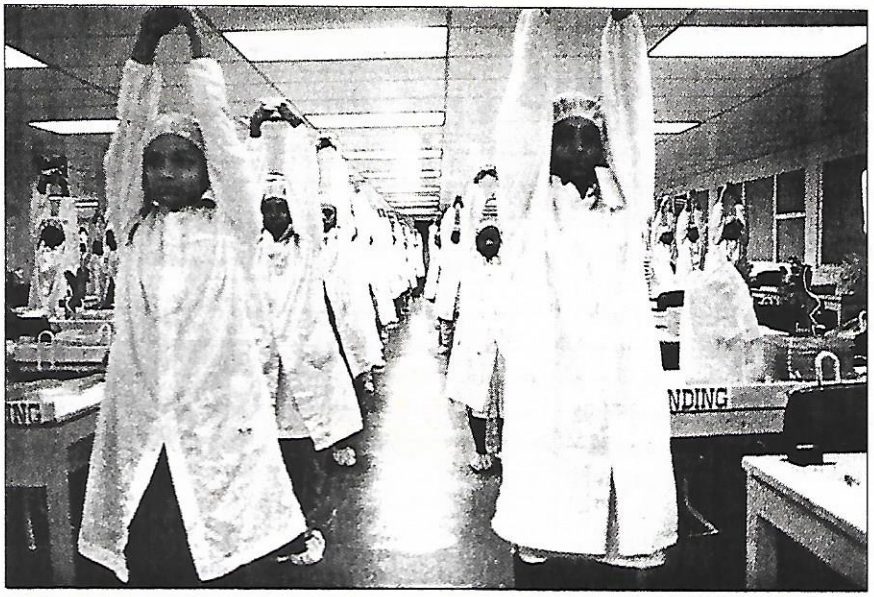
M.L. Tandon is India's [illegible]-largest computer hardware maker - and its largest exporter. Tandon's cutting edge is low-cost labour, rigorous quality norms, and support from associate firms overseas, say **Vanita Kohli** *and* **Jayanta Ghosh.**

Photographs by Vilas Kalgutker

It's a daily occupation. For 15 minutes, workers at Tandon Group's SEEPZ (Santa Cruz Electronics Export Processing Zone) unit in Bombay are taught *yoga* to relieve tension. The reason is obvious: day after day, men and women use microscopes to wind a copper wire - as thin as a human hair - 34 times around a ferrite core which is eventually mounted on a head-stack (a key component that performs the read-and-write function on the hard-disk drive of a personal computer). Nothing could be a more apt personification of CEO Manohar Lal Tandon's business strategy - employing India's traditional advantage, low-cost labour, in a hi-tech field.

The payoff: Tandon is India's leading exporter of computer hardware and its third largest manufacturer (behind the Rs 340-crore HCL-HP and the Rs 337-crore Wipro Infotech). In 1993-94, out of a national total of Rs 427 crore, the Bombay-headquartered infocorp alone exported Rs 208 crore worth of head-stacks, power supplies, voice coils, motors, tape heads, and electronic sub-assemblies (see *Tandon's Product Portfolio*).

On the export front, Tandon has made a complete sweep. The first three among India's top 10 hardware exporters belong to the Tandon stable: Ultra Tek Devices (1993-94 exports: Rs 72.8 crore), Advance Technology Devices (Rs 67.4 crore), and Tancom Electronics (Rs 46.9 crore).



A hi-tech break: workers at Tandon's SEEPZ unit at yoga

Group turnover has seen a jump of 39%, from Rs 159.7 crore (1992-93) to Rs 221.4 crore (1993-94), and Tandon - who has manufacturing facilities in Nasik, Indore, Madras, Delhi, besides Bombay (all independent profit centres) - expects to breach the Rs 275-crore mark in 1994-95. His clientele includes computer biggies like IBM, Western Digital, Seagate, Compaq and Olivetti.

In the process of commissioning units in Sri Lanka (for computer power supplies) and Madras (for hard-disk drives), Tandon has signed a joint venture agreement with America's Kollmorgen. On the anvil is an increase in capacities by 50% in the next five years. "I have achieved more than what I desired," says the 55-year-old Tandon. Apparently, though, that's not

Tandon's product portfolio

Customers like IBM, Compaq, Seagate, and Olivetti means quality is paramount. "These days, nobody negotiates on quality; it is taken for granted," says Golden Systems' J.J. Shrikhande. A particle of dust on a head-stack can cause a hard disk to crash; so products have to be manufactured with utmost care. Here's a peek at the 10 Tandon products:

Switch-mode power supplies (SMPS): They supply power to personal computers. An SMPS is fitted inside the computer in the form of square cabinets.

Winchester head-stacks: Headstacks perform the read-and-write function on a hard disk; Winchester is a brand-name. A Winchester disk drive is the most significant development in disk technology in recent years. The Tandon Group's new plant in Madras will export Winchester head-stacks from this year.

DC-to-DC convertors: They are used to stabilise voltage in a personal computer.

Head Gimbal Assembly: This is a type of head-stack.

Voice coils: These components are used in voice-activated computers.

Stepper motors: They are responsible for the rotation of the floppy

Brushless motors: They are a type of stepper-motors.

Tape-heads: They are used on cartridges, which are used to store back-up data on huge systems.

Electronic sub-assemblies: These include various cards which perform the control function on the hard disk drive.

Computer systems: Their exports started in 1989, but were discontinued in end-1992.

enough: "The world market has grown enormously - and our share is very low." Worldwide, the computer components industry is worth $275 billion (Rs 852,500 crore) and is slated to hit $295 billion (Rs 914,500 crore) in 1994-95. India's share of Rs 427 crore is measly indeed. Tandon's target for 2000: Rs 1,000 crore.

That's big time for the boy from Barnala (Punjab), who did his MS in electronics from Purdue University, USA. In 1963, Tandon joined IBM and also enrolled on a part-time MBA course at the University of Santa Clara. While working on a thesis - which involved charting his future path - he realised that if he stayed on in the US, he would be only an employee, but if he went back to India, he could have a business of his own.

Thoughts of home-coming lingered during the 15 years he spent at IBM. When he finally returned home in 1978, computers seemed to be the most logical business to enter. Those days, computers were "mystical, expensive and hard to use", says Tandon, who managed to rope in a lot of IBMers in his startup. The goal was clear-cut: Tandon wanted to be a net exporter - not a licensee or a representative for any multinational.

The hotshots

India's top 10 exporters in 1993-94

Ultra Tek Devices	72.8
Advance Technology Devices	67.4
Tancom Electronics	46.9
Silicon Valley Technology	42.3
Neptune Computers	16.4
ICIM	13.8
Wipro Infotech	5.2
Ratan Exports	4.6
Melstar Exports	2.5
Modi Olivetti	2.5

Note: The first three are Tandon firms
All figures in Rs crore; Source: ESC

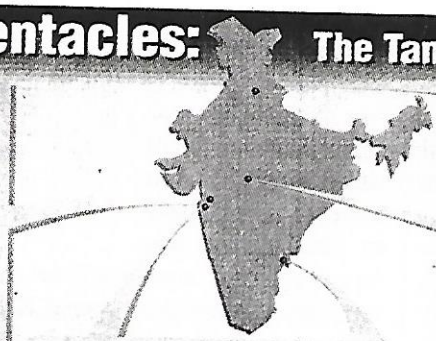
A company managed by his brother Jaggi Tandon in the US, Tandon Corporation, had developed a low-cost recording head which showed tremendous export potential. With Rs 25,000 in his pocket, Tandon set up shop in SEEPZ and started manufacturing floppy heads, followed by cable harness units and stepper motors.

Reminisces Tandon: "In 1981, when IBM launched the first personal computer, every head that went into the drive was made in Bombay." The first year saw Tandon notching up sales of more than Rs 40 lakh. But not everything was hunky-dory - infrastructure, bad today, was worse then. No direct-dialling to the US meant a whole day's wait for a booked call. Success came from Tandon's deep understanding of the mechanics of the computer components market, his ability to surmount problems in doing business in India, and low-cost labour, which Tandon leveraged well.

A skilled labourer in India earns around $80 (Rs 2,480) per month less than a dollar (Rs 31) a day - the figure was even lower when Tandon started his business. It made sense to make parts with a very high labour content: switch-mode power supplies, add-on cards, head-stacks, stepper motors, and voice coils. Integrated circuits or microprocessor chips need capital inputs which are prohibitively expensive.

But the benefits of cheap labour are neutralised by low productivity. Productivity in Taiwan and Singapore, India's main competitors, is estimated to

Spreading its tentacles: The Tandon Group's operations in India



BOMBAY

Tancom Electronics
- Head stack assemblies
- Voice coils
- Tape heads

Tandon Motors
- Stepper Motor
- Brushless DC Motor

Eastern Peripherals
- Floppy Disk drives
- Computer system
- Electronic sub-assemblies

Ultra-Tek Devices
- Switch Mode Power Supply
- DC to DC convertor
- AC adaptor

Advance Technology Devices
- Head stack assembly

Golden Computers
- Computer systems

NEW DELHI

Tancom Electronics, Nepz
- Voice coils

NASIK

Memory Electronics
- Voice coils
- Floppy disk drives
- Computer systems
- Power supplies

MADRAS

Ultra-Tek Devices
- Switch mode power supply
- DC to DC convertor
- AC adaptor

Modular Electronics
- Head stack assembly
- Voice coils

INDORE

Ideal Electronics
- Voice coils
- Power supplies
- Computer Systems

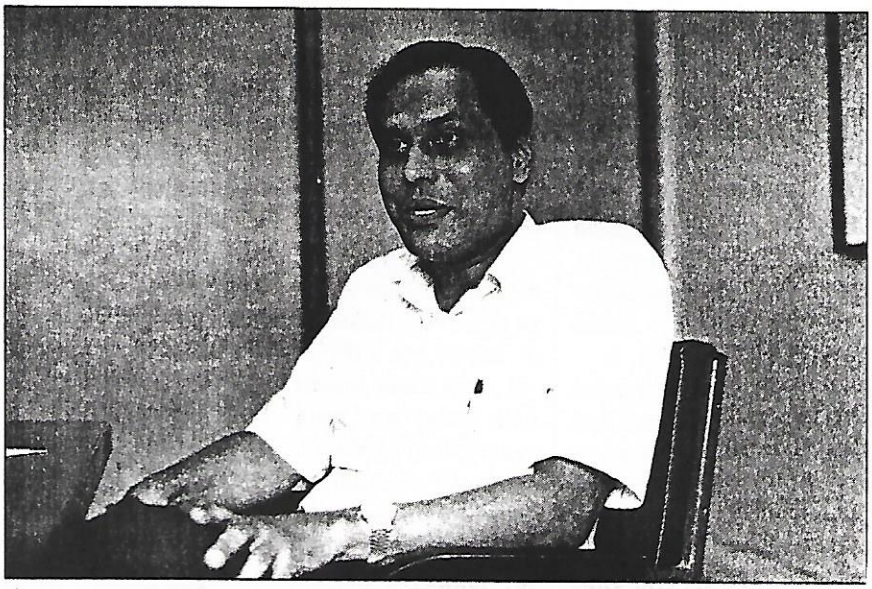
be 50% more than anything that comes out of SEEPZ. "Although India's skilled labour is one of the best in the world, productivity in Singapore and Taiwan is higher because of better infrastructure," says D.K. Sareen, chief of export promotion and market development, Electronics and Computer Software Export Promotion Council (ESC).

Also, the number of rejects, whose cost has to be absorbed by the good pieces, is much higher. Reject rates abroad are a fourth of those in India. Says J.J. Shrikhande, vice-president, worldwide manufacturing operations, Golden Systems (an associate company): "IBM and Motorola talk of zero defect, a concept which Indians find very difficult to digest. The concept is the key to mass manufacturing."

A high defect rate translates into more rework, higher cost, low productivity - and more scrap. With an estimated defect rate of 500-1,000 ppm (parts per million), the Indian hardware industry has a long way to go. The domestic industry sells rejects in grey market areas like Lamington Road in Bombay while pushing costs of good pieces by 10-20%.

"Compared to international standards, productivity is the lowest in India. That is why training is important," says Tandon. True, since computer components is a volume-driven business. More than 85% of Tandon's 5,300-strong workforce comprises women between 18-25. "The manufacture of components requires deft fingers," says Shrikhande. "Men get tired of repetitive work; women are able to perform better. This is a worldwide phenomenon."

Most of the women employed by Tandon have completed their 12th standard. Quality and cleanliness are drilled into them during training sessions which last for one-three months. It's important, since most foreign buyers do check out whether the systems are in place. Take head-stacks, for instance: they have to be fitted on hard disks in an absolutely clean environment. Advanced Technology Devices, a group company which makes heads, has rooms with 100k (k=1,000), 10k and 1k particles of dust per cubic foot - all conforming to international standards. Interestingly, the human skin alone sheds 100k particles of dust every minute. So, workers wear gowns, caps and use paper covers on their feet. Customers, say company officials, have actually measured the presence of dust themselves; to reassure them further, Tandon is awaiting the ISO 9000 certification.



Shrikhande: "The entire country should be turned into an EPZ"

A vendor should be able to anticipate new technologies.

"Worldwide, capital equipment and the cost of capital is the same. The real edge lies in people: if you have more people, you have a bigger competitive advantage. If we keep training workers, then labour will continue to be an advantage," says Tandon. "The advantage will last for the next 20 years. I don't think there is any region in the world, apart from India and China, where labour comes cheap," says Shrikhande.

Although vendors like Tandon thrive on very high manual inputs, they are hardly immune to the swings of technology. The average product life cycle (PLC) in the computer hardware industry is a low 6-12 months; in certain categories, it is just three months. Not only should a vendor be able to supply the latest component, he should be able to anticipate new technologies.

Normally, a product takes a couple of weeks to deliver. A new product takes between two and four months - and if the design is obsolete by then, too bad. It means that capital investment, which can run into crores of rupees, has come to nought - an absolute no-no for an exporter like Tandon, who sells half his output in the low-margin, high-volume markets of the US and Canada. Explains Tandon: "In the electronics industry, you cannot export anything that is obsolete. If our customers have the latest products, we have to supply them with the latest components."

It is not automation but the degree of automation that is important, says the CEO. "This can vary as and when the product demands it. My margin per

piece may go down but the margin per factory increases." Adds K.C. Misra, general manager, Advanced Technology Devices: "A new plant takes six months to design and another six months to commission; by that time, the technology could change. We have to keep in constant touch with world trends." For instance, if a new type of head-stack is to be made, new inputs have to ordered, skills have to upgraded, and suppliers have to be informed of the volumes. Besides, the upgradation process requires new people, more area - and larger capacities. The company's product range - which encompasses 10 different products - magnifies the problem.

Technology should put any hardware exporter out of business but for cheap labour and - as the Tandon success proves - location. As many as seven Tandon units (see *Spreading Its Tentacles*) are in export processing zones (EPZs) - four in Bombay, two in Madras, and one in Noida, near Delhi. Consider the facts: in 1993-94, more than 90% of India's hardware exports came from EPZs. Operating out of an EPZ has advantages that would make a domestic producer go green: duty-free imports of components which, otherwise, attract a duty of 60-80%; a fourth of the output can be hawked in the domestic market at half the import duty provided there is a minimum value addition of 15%. In 1993-94, the domestic market accounted for around 6% (Rs 13.8 crore) of Tandon's sales.

The group's value addition ranges between 25-30%; import content is as high as 70-75%. "We are forced to import since most inputs cost more than double in India," says Shrikhande. Take copper wire, a key head-stack input. Although freight accounts for half the wire's cost, imports are by far cheaper.

The group's value addition is about 25-30%; import content is a high 70-75%.

Helping the Indian outfit in its overseas thrust is associate company Tandon Associates. The US-based company keeps the Indian manufacturing base up-to-date on the world's leading personal computer market, Tandon's largest consumer. "Earlier, vendors used to ask for time to deliver - and then jack up capacities. Now, expansions have to be proactive," says Misra. There is another advantage in having big brothers overseas: today, most research and development (R&D) is done by US-based companies run by his brothers, who also provide the marketing support.

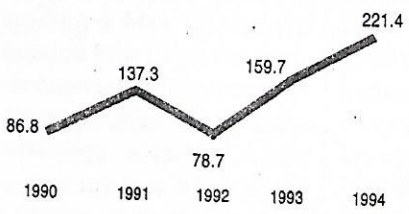
Marketing is one of the biggest problems faced by a hardware exporter, says ESC's Sareen. Till two years ago, around 10,000 Tandon computers were being exported to Europe every month but the cost of servicing and marketing proved prohibitive. (Today, personal computers have taken the back seat; the bulk of the group's sales come from power supplies for personal computers and subassemblies which go into hard disk drives.) Associate companies in the US - which liaise with Tandon's customers and suppliers - keep in touch with market needs. More than a decade of interaction with international buyers has helped build a strong customer base, Tandon's biggest plus, says Sareen. The Tandon Group, says Shrikhande, operates under four pressure points: quality, quantity, time, and design. A customer base of mostly OEMs (original equipment manufacturers) demands constant upgradation in all four areas.

A 16-year headstart should have helped Tandon to sell both his name and India's. Only the first task appears to have been accomplished in unique fashion. Tandon still has problems with the other: if something goes wrong on the buyer's production line, says a company official, the first suspect is the component with the made-in-India label. There are more woes: the recent plague scared away foreign technologists, who were setting up a clean room to make heads. Top off such scare with airline and dock strikes, riots and bomb blasts, and the recipe for disaster is complete.

In a cut-throat export market operating on slim margins, you cannot afford to goof up: one mistake, and out goes the customer. Says Misra: "We're low on margins and high on technology - and risk. It explains why few people enter this business. In the highly competitive environment, customers have no special love for Tandons or India or Singapore."

High-powered hi-tech

Sales profile of the Tandon Group

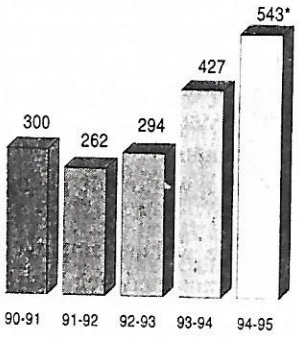


All figures in Rs crore; Source: company data

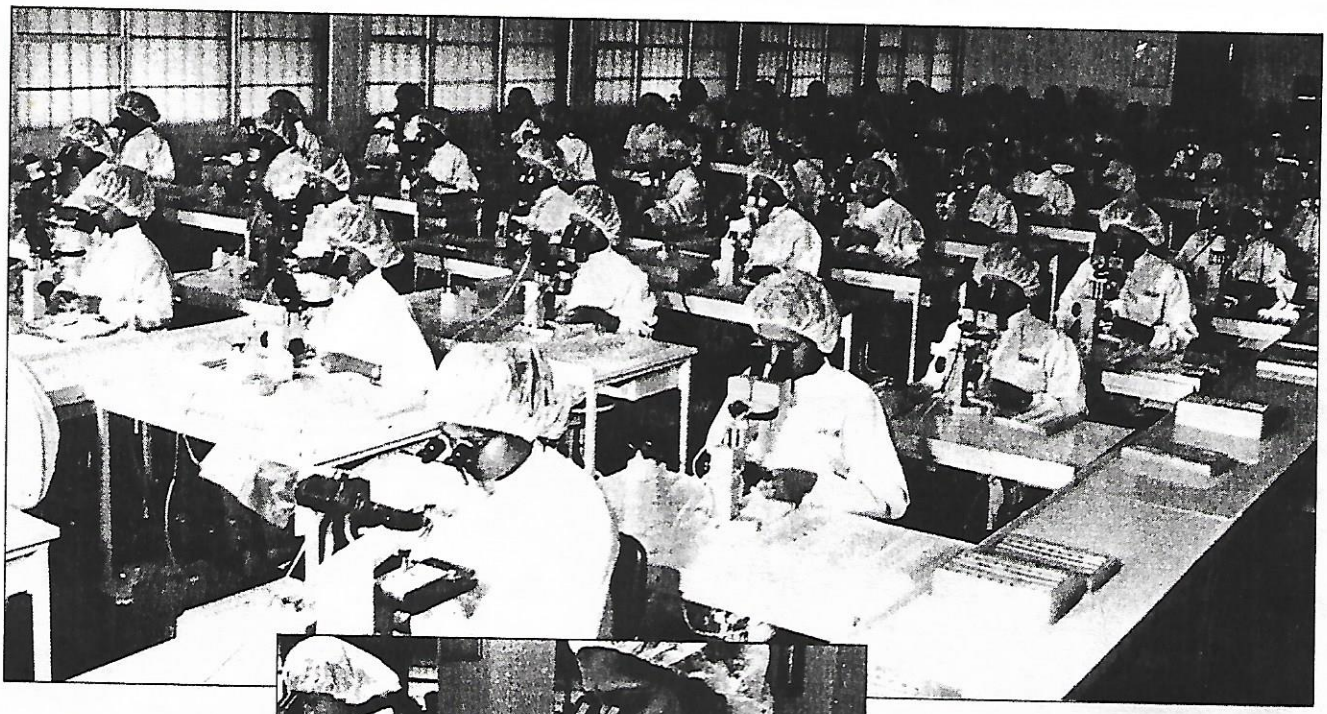
The poor performance in 1991-92, when sales dropped by over 40%, resulted from import restrictions. In fact, even national exports dipped, by over 12%

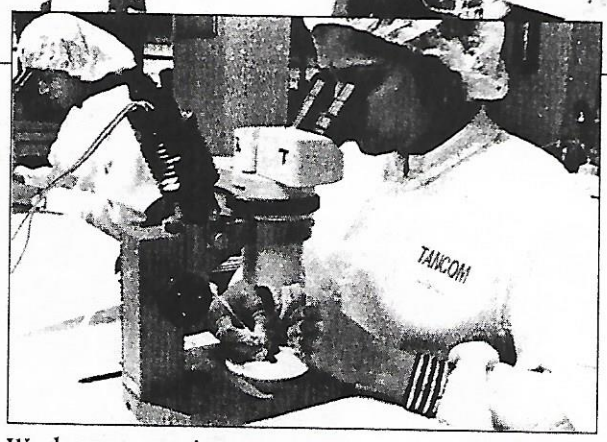
Hardware highs

India's exports of computer components



*Projections ; All figures in Rs crore; Source: ESC





Workers use a microscope to wind a copper wire - as thin as a human hair - around a ferrite core

If Tandon is on par with competing nations like Taiwan and Singapore on a cost-to-cost basis, he loses out on infrastructure, which determines deliveries. Importing takes over four weeks, clearances take forever, the customs is shut on Saturdays, and then there are those numerous holidays: these are problems all exporters have to deal with. "For every order, 40 different copies have to be filled in. Our customers don't understand all this," says Shrikhande.

Consider a nightmare scenario: the raw material ordered from abroad is bad. But sending it back takes more than two months! What does the company do? "In order to meet my commitments, I have to carry six-to-eight weeks of inventory, when the whole world is moving towards JIT (just-in-time). My foreign counterparts keep about six-to-eight days of inventory. We are losing to China in infrastructure and speed," says Tandon. He recounts the time when he took a consignment to the airport in his car to make it reach on time.

Compared to India, Sri Lanka is far better, says Shrikhande, who was involved in setting up a plant on the island. "There is no export zone in Sri Lanka; the entire country is an export zone. You don't have to pay any customs duty. The whole of India should become an EPZ." It explains why Golden Systems, an associate company, was set up in the island country to manufacture switch-mode power supplies. Besides, a shipment takes 10 days to reach the US from Sri Lanka compared to five weeks from India. "If the Indian industry does not become quality-, cost- and delivery-conscious within 10-15 years, it will be wiped out," says Shrikhande.

For the present, the hardware industry is simply bullish: exports jumped from Rs 274 crore (1992-93) to Rs 427 crore (1993-94). And the ESC expects exports to touch the Rs 543-crore mark this year. Much will depend on EPZ units, which account for the bulk of India's exports. The wafer-thin margins of this contract manufacturing business are already under pressure. The difference between India's and the world's prices ranges between plus or minus 5%, says Tandon. China, Malaysia, and Vietnam - Tandon considers them to be serious threats to India - are gradually closing in.

Is there any threat from within to India's leading hardware exporter? "We have done it for the last 15 years - it shouldn't be a problem for the next five years," says Tandon, who obviously derives his strength from people. Worldwide, companies are slashing jobs to cut costs and sharpen the competitive edge. An entrepreneur who uses manpower to create a Rs 221-crore group in a hi-tech market - and plans to continue doing so - comes as a refreshing change.